The rates of descent recorded in 401 runs to drop fire retardant are shown in figure 4. Approaches for the drops varied from flat runs having zero rates of descent to extremely steep runs in which descent rates of up to 9,000 feet per minute were recorded. The average rate of descent was 2,483 feet per minute. Discussions with personnel flying the retardant bombers indicate that descent rates were affected to a large extent by the physical characteristics of the terrain near the fire. Fires along hill tops or on ridges could be approached for the retardant drop at relatively shallow angles, whereas, steep approaches were necessary to reach fires along sides of ridge walls or on canyon bottoms. Based on the data of figure 4, it was calculated that 30 percent of the runs made by retardant bombers in mountainous areas can be expected to equal or exceed rates of descent of 3,000 feet per minute.

Figure 5 shows the airspeeds at the start of the highest descent rate in the retardant drop run and at pullout for 401 runs in which retardant drops were made. The results are shown as the summation of the number of drops made in each 5-knot increment of airspeed. The design flap speed, V_F, is also noted on the figure for comparative purposes.

Start speeds ranged from as low as 100 knots to as high as 211 knots. The low-start speeds were normally recorded during the second and succeeding retardant runs in a flight since the aircraft at such time was near the top of a climbing recovery from a preceding run and was, therefore, at a relatively low airspeed. The higher start speeds were usually experienced on the first run.

Pullout speeds ranged from 116 knots to 236 knots. The speed at the instant of pullout was dependent to a large extent on the type of approach required to reach the fire. Lower pullout speeds were usually recorded during flat or shallow approaches and the higher pullout speeds during steep approaches. The pilots indicated that approach speeds were sometimes controlled by using flaps as speed brakes in order to arrive over the drop area at the desired drop speed of 155 to 160 knots. Figure 5 shows that 105, or about 26 percent, of the pullouts from retardant drop runs were made at airspeeds of 160 knots or higher. Thirty-five of the pullout speeds and five of the start speeds exceeded the design flap speed; however, since the VGH record does not indicate flap position, it cannot be assumed that flaps were used during these runs.

The frequency distribution of the maximum positive and negative load factors recorded during 1,175 retardant drops is given in table III. Figure 6 shows a plot of these maximum load factors for each airplane as a summation of the number of runs for each 0.1 value of load factor. The aircraft design maneuver limit and ultimate load factors are also shown. A comparison of the load factors experienced by the two airplanes indicates that values up to 3.1 were experienced more often by airplane 2 than by airplane 1. Since both airplanes were flown by each of four pilots (ages 35 to 55) no reason for the difference in load factors could be attributed to differences in piloting techniques. Because the differences are relatively small, however, they are not considered significant.